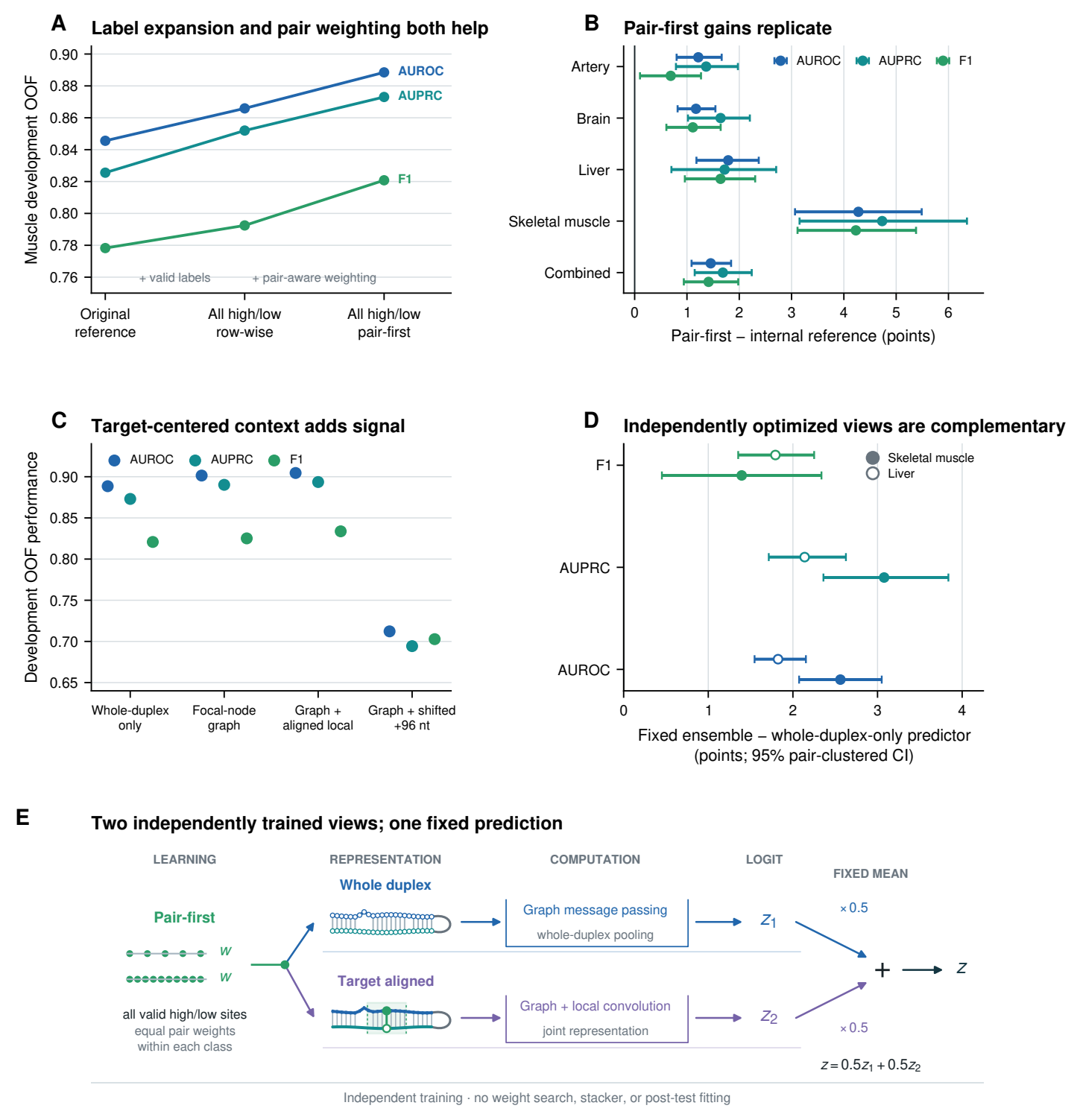
A
Label expansion and pair weighting both help
Muscle development OOF
0.90
0.88
0.86
0.84
0.82
0.80
0.78
0.76
AUROC
AUPRC
F1
+ valid labels
+ pair-aware weighting
Original reference
All high/low row-wise
All high/low pair-first
B
Pair-first gains replicate
Artery
Brain
Liver
Skeletal muscle
Combined
AUROC
AUPRC
F1
0
1
2
3
4
5
6
Pair-first – internal reference (points)
C
Target-centered context adds signal
Development OOF performance
AUROC
AUPRC
F1
0.90
0.85
0.80
0.75
0.70
0.65
Whole-duplex only
Focal-node graph
Graph + aligned local
Graph + shifted +96 nt
D
Independently optimized views are complementary
F1
AUPRC
AUROC
Skeletal muscle
Liver
0
1
2
3
4
Fixed ensemble – whole-duplex-only predictor
(points; 95% pair-clustered CI)
E
Two independently trained views; one fixed prediction
LEARNING
REPRESENTATION
COMPUTATION
LOGIT
FIXED MEAN
Pair-first
w
w
all valid high/low sites
equal pair weights
within each class
Whole duplex
Graph message passing
whole-duplex pooling
z_1
Target aligned
Graph + local convolution
joint representation
z_2
×0.5
×0.5
+
z
$z = 0.5z_1 + 0.5z_2$
Independent training · no weight search, stacker, or post-test fitting